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Project 1

PBHLT 7101

3/3/2026

Introduction

The purpose of this analysis is to explore relationships between various anthropometric measures. Our primary measures of interest are measures of stature, weight, and BMI. The data source comes from the 2012 Army Anthropometric Survey (ANSUR II). This is a survey that was created by the United States Army with the stated goal of collecting physical characteristics of service members that would be representative of, “the total US Army force to include the US Army Active Duty, Reserves, and National Guard”. They collected data for more than 90 physical metrics, and 15

demographic or administrative metrics. This included measures of overall height and weight, size measured for specific areas of the body, and distances between specific parts of the body.

Analysis Plan

1. Organize Workspace
    a. Create new folder for this project
    b. Download data and move it to a data folder within the project folder
    c. Create project files including do file and report doc
    d. Create file header for do file
2. Clean Data
    a. Read through codebook provided for special values that will need to be recoded for
    b. Recode data based on codebook using special values
    c. Explore data, look for anomalous values
    d. Document any anomalous values
    e. Flag anomalous values via a flag variable for each column with anomalous values
    f. Correct anomalous values
    g. Providing documentation that the coding is correct
    h. Identify a primary key
    i. Check duplicate observations
    j. If duplicate observations are found, prioritize the cleanest record (e.g. fewest suspect values), if tied, then keep the first observation
    k. Label all variables and values of categorical variables.
    l. Remove subjects under the age of 18
    m. Convert the measurements to cm from mm
3. Add New Variables
    a. Continuous BMI
    b. Categorical BMI using standard definitions of obese, overweight, normal weight, and underweight
    c. Season the measurements were made
    d. Gender flag
    e. Preferred hand flag
    f. Categories of body types based on height, weight and other measurements
4. Data Descriptions
    a. Generate statistics describing the anthropometric measures of the sample
    b. Summarize measures for the entire sample in a table. For clarity organize the table by body part
    c. Note the number of missing values and suspicious values for each variable
    d. Create a figure that shows the differences in % of total height attributable to the height to hip by sex
    e. Add figures to report

    f. Describe results from part d in the report
    g. Create a data quality assessment for the report analyzing the data quality issues
5. Data Analysis
    a. Assess the level of correlation between the measures of stature
    b. Identify the two measures that are most highly correlated and provide a figure that demonstrates the relationship between these measures
    c. Assess the level of correlation between overall height and the other measures of stature, stratified by sex.
    d. Compare the self-reported weight to the measured weight including the distribution of the difference. Use a histogram for this
    e. Show the extent to which the difference between reported and measured weight varies by sex. Use a box plot to show this
    f. Compare the self-reported height to the measured height describe the distribution of the difference. Use a box plot for this

Data Preparation and Quality Assessment

The documentation provided for the data offered pre-defined values to recode using special missing values, see the appendix for special value mappings, so re-coding these values became the first step in data preparation. Along with re-coding these values, subjects under the age of 18 were removed from the dataset to align with the study requirements. After this initial data cleaning, a quick data exploration revealed anomalous values in multiple variables. There were several physically impossible measurements, specifically *thumbtipreach* had values ≥ 6000 and *weightlbs* had measurements of 0. Observations with these measurements were flagged with corresponding flag variables, then re-coded as *.d* in the analytics data set. Categorical inconsistencies were also observed, this included values of *dodrace* responses that were not valid according to the provided documentation and the correction of spelling errors within the *writingpreference* field. To handle incomplete information, blank string entries were assigned a missing indicator, while numeric outliers in *subjectnumericrace* were flagged for further review. Following these adjustments, all the values fell within plausible ranges.

Along with anomalous and missing values, there was a significant number of clearly duplicate values. The data set did not include a unique subject identifier, so a surrogate key was generated using values that were considered stable and the date of observation. This approach balanced data integrity without being overly sensitive and removing observations that we could not guarantee were duplicative. The survey was conducted with the goal of being unique at the subject level, but that level of granularity cannot be guaranteed without an official subject ID, so the results of this study should be interpreted at the subject-observation date granularity. Table 2.1 depicts anomalous and missing values by variable after completing data cleaning, transformation, and labeling.

Table 2.1 Missingness by variable

Variable Name	*Non-Missing*	*Missing*	*Not Recorded*	*Refused*	*Unknown Reason*	*Invalid/ Implausible Value*	*Blank String*
subjectnumericrace	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
dodrace	6065 (99.7%)	18 (0.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	18 (0.3%)	0 (0.0%)
age	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
weightlbs	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)
heightin	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
thumbtipreach	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
span	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
footlength	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
kneeheightmidpatella	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
waistheightomphalion	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
functionalleglength	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
cervicaleheight	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
trochanterionheight	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
stature	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
waistcircumference	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
chestcircumference	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
bicristalbreadth	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
hipbreadth	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
hipbreadthsitting	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
weightkg	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)
date	6062 (99.7%)	21 (0.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
flag_subjectnumericrace_hi	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
ethnicity_miss	0 (0.0%)	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	4659 (76.6%)
pk_row	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
subject_key	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
handedness_cat	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gender_cat	6083 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
bmi	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
bmi_cat	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
month_obs	6062 (99.7%)	21 (0.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
season	6062 (99.7%)	21 (0.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
whtr	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
central_adip	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
ht_tertile	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
bmi_tertile	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
body_type	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
pct_hip_height	6082 (100.0%)	1 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)

Anthropomorphic Characteristics

Table 3.1 Summary stats by body part

Measure	*N*	*Mean*	*SD*	*Median*	*Min*	*Max*
Hip breadth, sitting (cm)	6,082	38.577	3.242	38.400	28.000	54.100
Waist circumference (cm)	6,082	91.453	11.417	90.700	61.100	137.900
Chest circumference (cm)	6,082	102.210	10.061	102.050	69.500	146.900
Bicristal (pelvic) breadth (cm)	6,082	27.474	1.921	27.400	19.700	36.200
Hip breadth (cm)	6,082	34.841	2.530	34.700	26.400	47.300
Arm span (cm)	6,082	176.390	11.084	177.100	132.300	212.100
Standing height (cm)	6,082	171.445	9.001	171.900	140.900	199.300
Knee height at mid-patella (cm)	6,082	47.550	3.307	47.600	34.600	61.200
Cervicale height (cm)	6,082	147.753	8.432	148.400	118.400	173.800
Trochanterion height (cm)	6,082	88.273	5.439	88.200	66.000	109.500
Waist height at omphalion (cm)	6,082	103.155	6.272	103.300	80.500	124.500
Functional leg length (cm)	6,082	110.207	6.854	110.400	83.500	131.600
Foot length (cm)	6,082	26.305	1.739	26.400	19.800	32.300
Forward arm reach to thumb tip (cm)	6,082	78.977	5.382	79.200	60.200	99.800

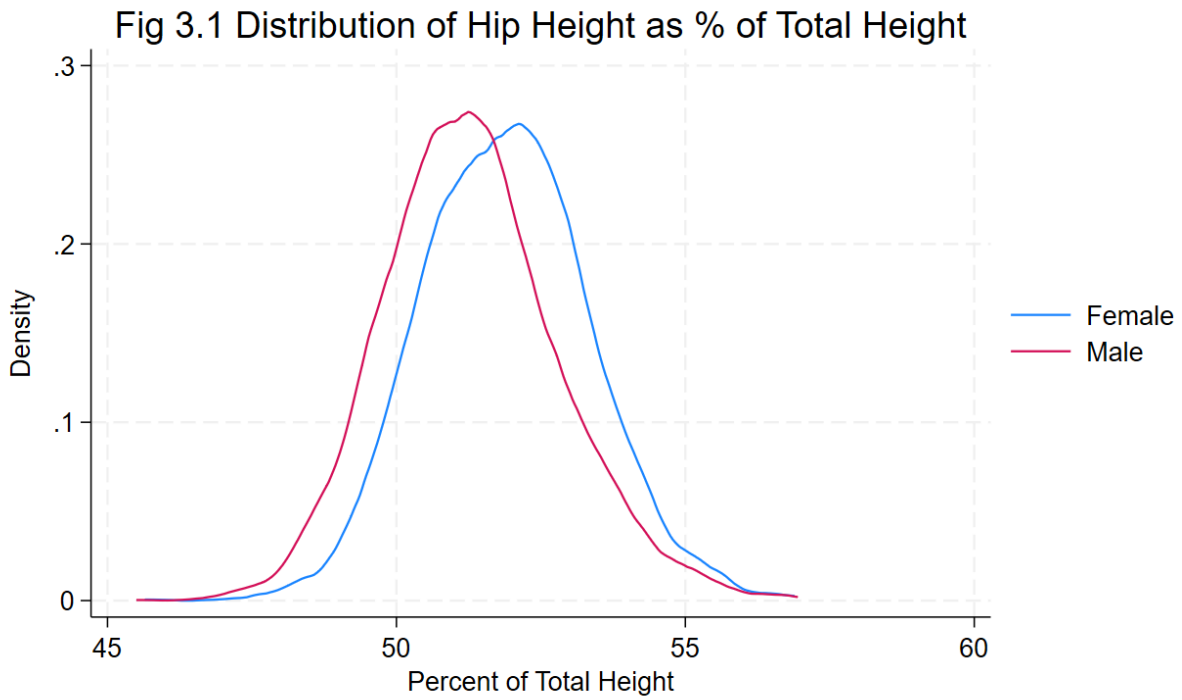


Figure 3.1 describes the distribution of subjects' hip height as a percentage of their total height by gender. The distributions appear to be approximately normally distributed for both men and women, with men shifted towards the right compared to women. Women also seem to have a lower spread in the distribution relative to men, meaning they are clustered closer to the center of their distribution. Men have a mean of approximately 52.5%, meaning all parts of their body up to their hip make up a larger proportion of their height than everything above their hips. The mean value for men is slightly higher than the mean value women, who have a mean percentage of roughly 51%.

Analysis

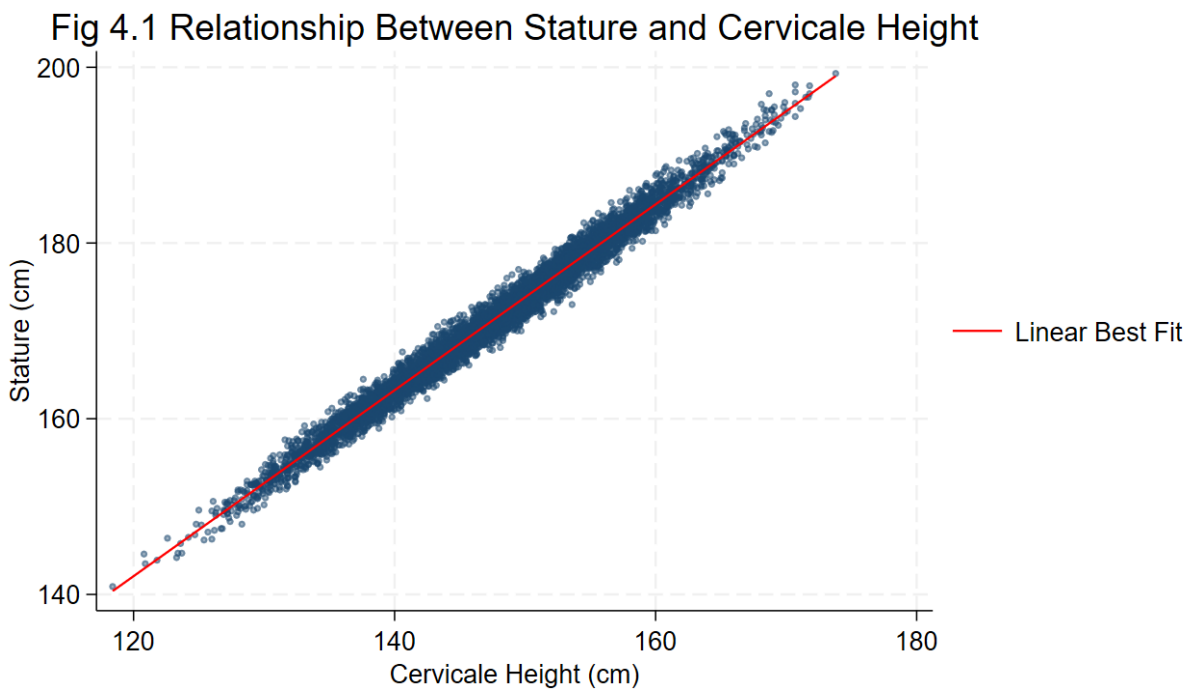


Figure 4.1 shows the relationship between stature (overall height) and cervicale height, with both being measured in centimeters. A subject's cervicale height had the highest correlation with their stature of any measure in the data set. This makes a lot of sense considering the measurement from the ground to a subject's cervicale makes up the largest proportion of the subject's height of any measurement. The correlation coefficient was found to be approximately .998, showing near perfect positive correlation. This strong correlation is depicted in Figure 4.1 with the data being clustered extremely close to the line of best fit and following the trajectory nearly perfectly without any noticeable outliers.

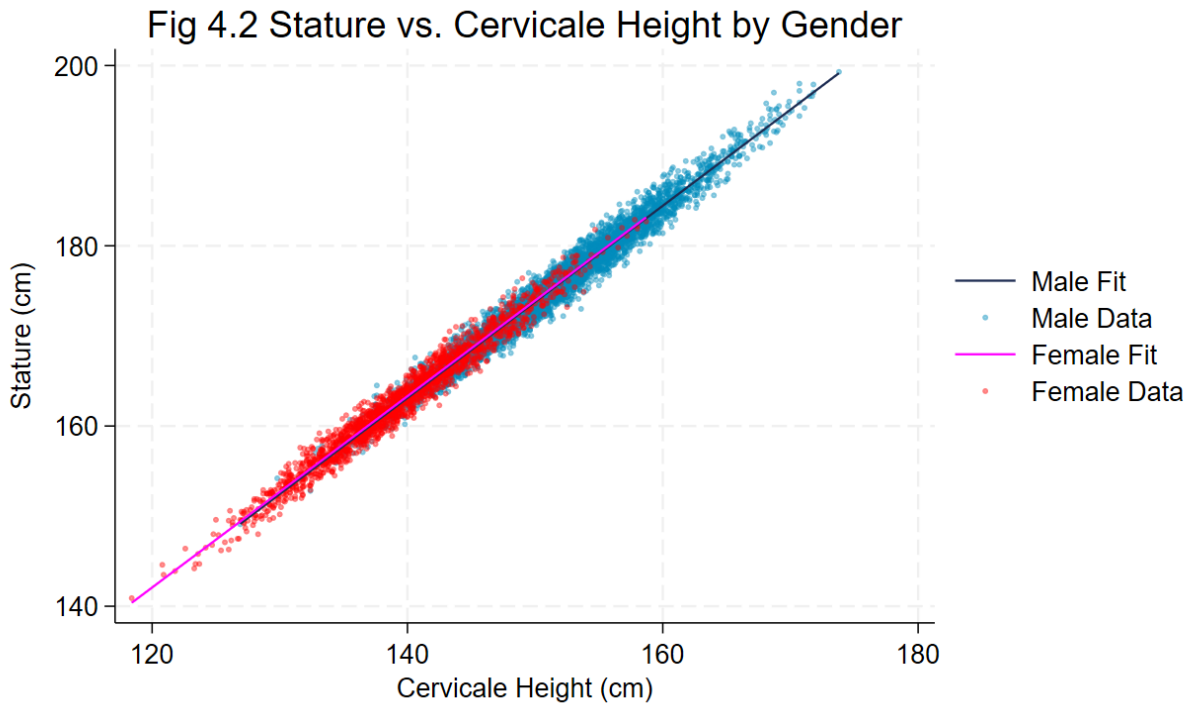


Figure 4.2 shows the same thing as figure 4.1, except the data is stratified by gender. Female subjects are represented by red dots and males in blue, and separate lines of best fit for each strata. The cervicale height had the largest correlation coefficient for both groups, with both of them having a coefficient larger than .98. The separate lines of best fit show that the linear relationship is identical between both groups, as they share the same slope and nearly perfectly overlay one another. The only noticeable difference between the two groups is the female subjects have a lower mean cervicale height and overall height.

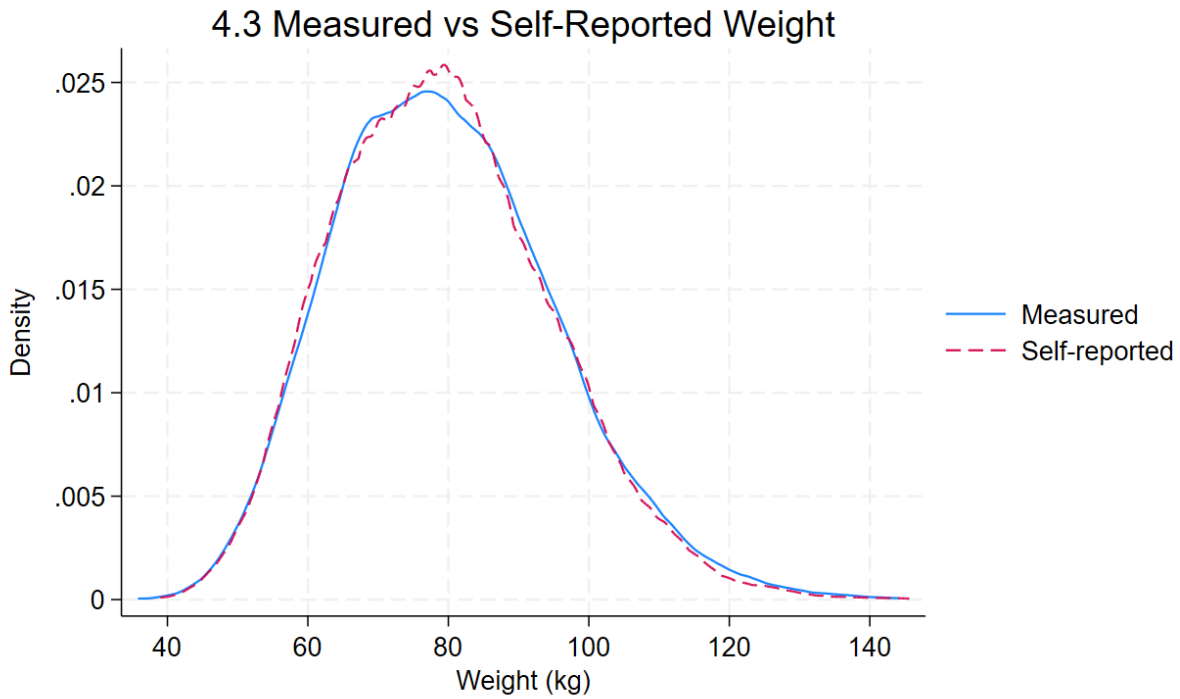
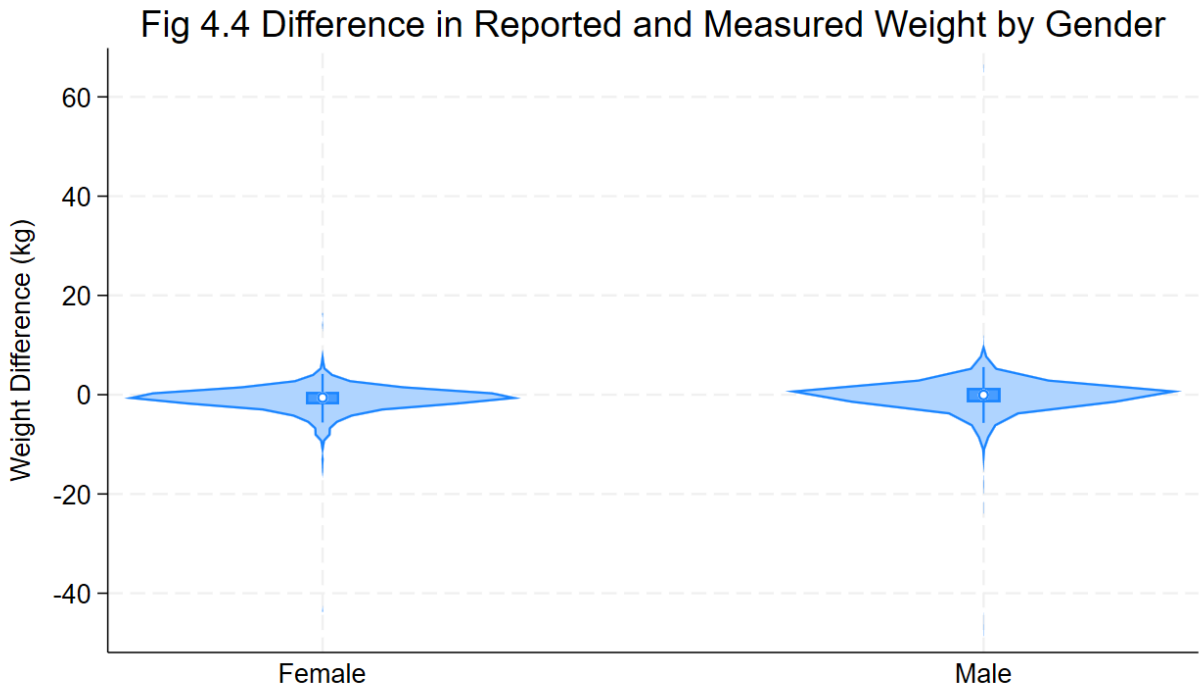


Figure 4.3 shows the distribution of self-reported weight and the weight of the subject at the time of measurement. The data appears to be left skewed with a long tail towards the right. The measured and self-reported weights track pretty closely to one another except for at the center where weights around 79kg are overrepresented while the weights slightly under 79kg are underrepresented. This indicates that a large number of subjects over-reported their weights. 79kg corresponds to 180lbs in imperial units, so it is possible this discrepancy is due to subjects rounding up.



These plots depict the distributions of difference in reported and measured weight by gender. This was calculated as *Reported – Measured*, so a positive value would mean that a subject overstated their weight, while a negative value would mean that a subject understated their weight. There appears to be a slight difference between the genders towards the center, with males tending to overstate their weight by a small amount and women tending to understate their weight by a small amount, however, both genders are centered very close to 0.

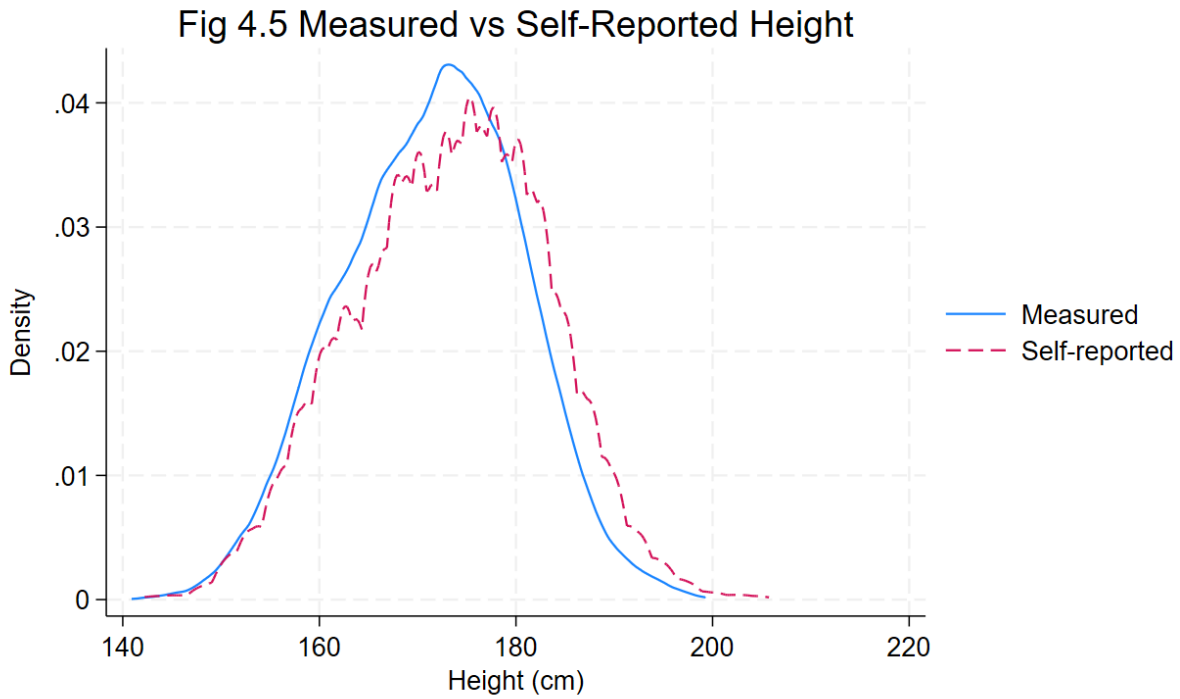


Figure 4.5 compares measured and reported height in subjects. The distribution of measured height is noticeably smoother than the reported height, this is most likely due to the reported heights being in inches and then converted to centimeters. There also appears to be a slight systematic bias towards subjects overstating their heights. We can see the densities towards the lower ends of the chart are lower for self-reported values than what the measured values indicate they should be while the densities for self-reported heights over 180cm are higher than what the measured values indicate.

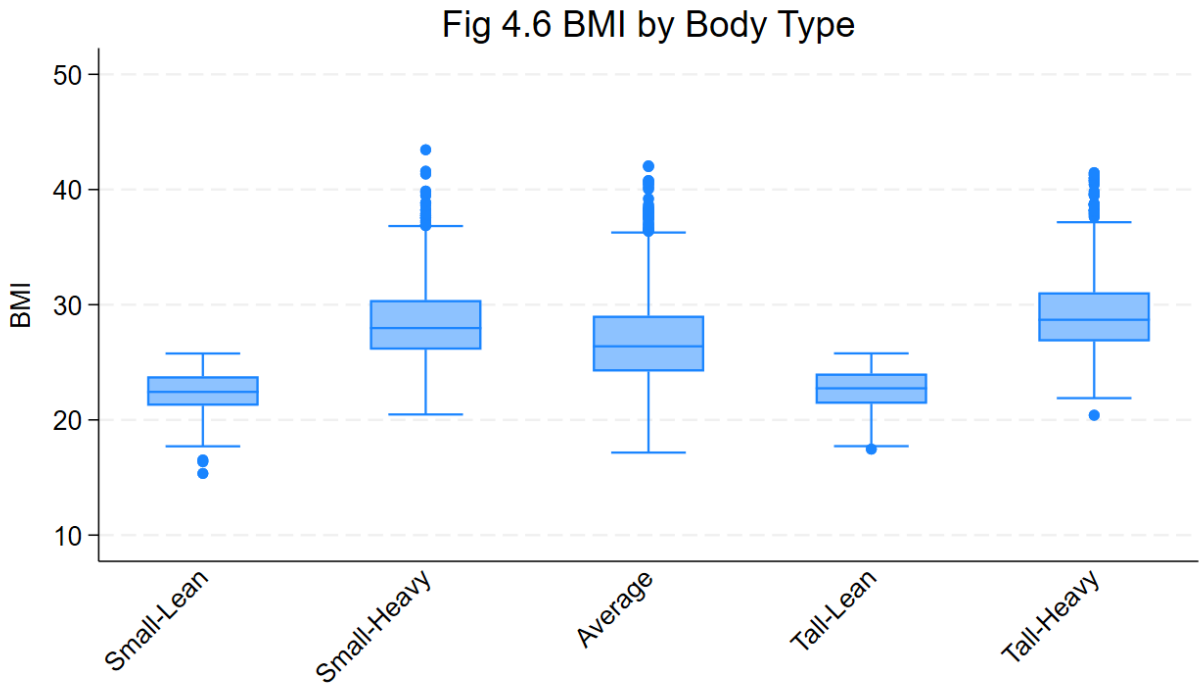


Figure 4.6 demonstrates these newly defined body types have distinct BMI distributions, with the "Heavy" categories showing significantly higher median BMI values and greater variability compared to the "Lean" and "Average" groups. Specifically, the median BMI for "Small-Heavy" and "Tall-Heavy" types sits near the 30 kg/m mark, indicating that these categories tend to capture individuals likely to be classified as obese. In contrast, the "Small-Lean" and "Tall-Lean" groups show much tighter distributions centered around the low 20s.

Appendix

```
*******************************************************************************
**                                                                           **
** Name: Travis Swiger                                                       **
** Date: 2/21/2026                                                           **
** Assignment: Project 1                                                     **
**                                                                           **
*******************************************************************************


// 0. Setup assignment environment, import and explore data

cd "C:\Users\TravisSwiger\Documents\homework\Data Mgmt\stata_files"

use "data/ansur2allV2.dta", clear


codebook

describe

summarize, detail


*******************************************************************************

// Section 1: Clean Data

*******************************************************************************


// A. Pre-defined tasks
```

```
// -77 = .a = not recorded

// -88 = .b = refused measurement

// -99 = .c = unknown missing

// Remove participants under 18

ds, has(type numeric)

foreach var of varlist `r(varlist)' {

   recode `var' ///

      (-77 = .a) ///

      (-88 = .b) ///

      (-99 = .c)

}


drop if age < 18 & age < .



*******************************************************************************

// B. Look for anomalous values

* Separate male and female as we should expect the ranges to be different between the two groups

ds, has(type numeric)

local numvars `r(varlist)'

tabstat `numvars', ///

   by(gender) ///

   statistics(min max p10 p25 p50 p75 p90) ///

   columns(statistics)
```

// Findings //

* weightkg: Max is over 800kg, setting highest plausible as 200kg

* heighin: Max is 94, setting max to 84in (7ft) as highest plausible

* subjectnumericrace: Flag values > 1000

* dodrace: replace missing with '.d'

* missing strings: define as ""

* handedness: Fix spelling

* weightlbs: Replace with missing ".d" if = 0

* thumbtipreach: Flagged values >= 6000 as data entry errors

* All other values appear reasonable

****

// D. Clean anomalous values

* For newly identified missing values:

* .d = numeric "invalid/should be missing" (e.g., weightlbs==0)

* .e = string "blank/empty response"

// Clean up numeric values

replace weightlbs = .d if weightlbs == 0 // weightlbs: 0 is impossible -> .d

replace weightkg = .d if weightkg >= 200 // Max weight > 800kg, -> .d

replace dodrace = .d if dodrace == 7 // Not a valid response according to documentation

replace heightin = .d if heightin > 84

replace thumbtipreach = .d if thumbtipreach >= 6000 & thumbtipreach < . // thumbtipreach: clearly impossible values, likely data-entry -> set to .d

```
// subjectnumericrace: suspected mis-coded / not race; do not recode automatically, flag for review

capture confirm variable subjectnumericrace

if _rc==0 {

   gen flag_subjectnumericrace_hi = (subjectnumericrace > 1000 & subjectnumericrace < .)

   label var flag_subjectnumericrace_hi "subjectnumericrace unusually high (review)"

}


// Clean up blank strings

foreach v in ethnicity component branch writingpreference installation {

   capture confirm variable `v'

   if _rc==0 {

      replace `v' = strtrim(`v')

      replace `v' = "" if `v' == ""   // ensures whitespace-only becomes ""


      * create a companion missing-indicator coded as .e for blank responses

      gen `v'_miss = .

      replace `v'_miss = .e if `v' == ""

      label var `v'_miss "`v': blank string response (.e)"

   }

}


// writingpreference spelling fix: "Either han" -> "Either hand (No preference)"

capture confirm variable writingpreference

if _rc==0 {

   replace writingpreference = strtrim(writingpreference)
```

```
   replace writingpreference = "Either hand (No preference)" ///
      if writingpreference == "Either han"
}

*******************************************************************************

// E. Identify Primary Key
* No subject ID included
* Need to use stable variables

* Always-unique row key
gen long pk_row = _n
label var pk_row "Primary key: unique row id in this dataset"

* Candidate subject key using stable variables
* ethnicity excluded due to number of missing values
* age included as it is stable for subject-date of measurement granularity

local stable gender dodrace stature footlength heightin
egen long subject_key = group(`stable'), missing
label var subject_key "Proxy subject id from stable fields"

*******************************************************************************
// F. Handle duplicates
```

```
* Expected unique at subject-date level
duplicates tag subject_key test_date, gen(dup_subjdate)
label var dup_subjdate "Duplicate count tag: subject_key x test_date"
gen byte flag_possible_dup = (dup_subjdate>0)
label var flag_possible_dup "Possible duplicate (subject_key x test_date)"

* Define the measurement set you care about for "exact duplicates"
local meas weightlbs heightin thumbtipreach span footlength kneeheightmidpatella ///
    waistheightomphalion functionalleglength cervicaleheight ///
    trochanterionheight stature waistcircumference chestcircumference ///
    bicristalbreadth hipbreadth hipbreadthsitting weightkg

* Tag exact duplicates (all meas identical within subject_key+test_date)
duplicates tag subject_key test_date `meas', gen(dup_exact)
label var dup_exact "Exact duplicate on subject_key+test_date+measures"

* Count missing among key measures (lower is better)
egen byte nmiss_meas = rowmiss(`meas')
label var nmiss_meas "Number of missing values across key measures"

* Sort within subject-date by completeness, then by pk_row for determinism
bysort subject_key test_date (nmiss_meas pk_row): gen byte keep_best = (_n==1)
label var keep_best "Keep-best record within subject_key x test_date"

* Drop exact duplicates
```

```
preserve
   keep if dup_exact>0 & keep_best==0
   save "proj1_duplicates_exact.dta", replace
restore
drop if dup_exact>0 & keep_best==0

* Drop the removed duplicates from main analytical dataset
drop if flag_possible_dup==1 & keep_best==0

display "Total obs: " _N
count if flag_possible_dup==1
display "Obs in duplicated subject-date sets: " r(N)

count if dup_exact>0
display "Obs that are exact duplicates: " r(N)

* After dropping, confirm uniqueness at intended grain
isid subject_key test_date


*******************************************************************************
// G. Label all variables and values of categorical variables

// Demographics and Administrative Variables
```

```
label var subjectnumericrace   "Subject numeric race code (original data field)"
label var dodrace              "Department of Defense race category"
label var ethnicity            "Self-reported ethnicity"
label var gender               "Gender"
label var age                  "Age at time of measurement (years)"
label var component            "Military component (e.g., Regular Army, Reserve)"
label var branch               "Military branch or occupational branch"
label var writingpreference    "Handedness / writing preference"
label var installation         "Installation where measurements were taken"
label var test_date            "Measurement date (string as recorded)"
label var date                 "Measurement date (Stata internal date)"
label var strdate              "Measurement date (cleaned string format)"

// Self-Reported Anthropometrics

label var weightlbs            "Self-reported weight (pounds)"
label var heightin             "Self-reported height (inches)"

// Measured Anthropometrics (recorded in millimeters unless noted)

label var thumbtipreach        "Forward arm reach to thumb tip (mm)"
label var span                 "Arm span (mm)"
label var footlength           "Foot length (mm)"
label var kneeheightmidpatella "Knee height at mid-patella (mm)"
label var waistheightomphalion "Waist height at omphalion (mm)"
```

```
label var functionalleglength  "Functional leg length (mm)"
label var cervicaleheight      "Cervicale height (mm)"
label var trochanterionheight  "Trochanterion height (mm)"
label var stature              "Measured stature / standing height (mm)"

label var waistcircumference   "Waist circumference (mm)"
label var chestcircumference   "Chest circumference (mm)"
label var bicristalbreadth     "Bicristal (pelvic) breadth (mm)"
label var hipbreadth           "Hip breadth (mm)"
label var hipbreadthsitting    "Hip breadth, sitting (mm)"

label var weightkg             "Measured weight (kg × 10; divide by 10 for kg)"

// Data Quality Flags and Derived Variables

label var flag_subjectnumericrace_hi ///
   "Flag: subjectnumericrace unusually high (review)"

label var ethnicity_miss ///
   "Missing indicator: ethnicity blank string (.e)"

label var component_miss ///
   "Missing indicator: component blank string (.e)"

label var branch_miss ///
```

```
   "Missing indicator: branch blank string (.e)"

label var writingpreference_miss ///
   "Missing indicator: writing preference blank string (.e)"

label var installation_miss ///
   "Missing indicator: installation blank string (.e)"

// Primary Key and Duplicate Management

label var pk_row       "Primary key: unique row identifier"
label var subject_key   "Proxy subject identifier (derived fingerprint key)"
label var dup_subjdate  "Duplicate tag: subject_key × test_date"
label var flag_possible_dup ///
   "Flag: possible duplicate at subject-date level"

label var dup_exact     ///
   "Exact duplicate across subject-date and measurement fields"

label var nmiss_meas    ///
   "Count of missing values across key measurement variables"

label var keep_best     ///
   "Indicator: retained best record within duplicate set"
```

```
**************** Label Categorical Values ****************

// Missing, Special, or Null Values

label define miss_ext ///
   .a "Not recorded (-77)" ///
   .b "Refused (-88)" ///
   .c "Unknown missing (-99)" ///
   .d "Invalid/implausible value (recoded)" ///
   .e "Blank string response (string missing indicator)", modify

ds, has(type numeric)
label values `r(varlist)' miss_ext
// dodrace
label define dodrace_lbl ///
   1 "White" ///
   2 "Black" ///
   3 "Hispanic" ///
   4 "Asian" ///
   5 "Native American" ///
   6 "Pacific Islander" ///
   8 "Other" ///
   .a "Not recorded (-77)" ///
   .b "Refused (-88)" ///
   .c "Unknown missing (-99)" ///
```

```
   .d "Invalid/implausible value (recoded)" ///
   .e "Blank string response (string missing indicator)", replace

label values dodrace dodrace_lbl
label var dodrace "Department of Defense race category"

**************** Encode Categorical Values ****************
// handedness
encode writingpreference, gen(handedness_cat)
label var handedness_cat "Writing preference / handedness (categorical)"

// gender
encode gender, gen(gender_cat)
label var gender_cat "Gender (categorical)"
label define gender_lbl 1 "Female" 2 "Male", replace
label values gender_cat gender_lbl

*******************************************************************************
// H. Convert the measurements to cm from mm
local mmvars thumbtipreach span footlength kneeheightmidpatella ///
        waistheightomphalion functionalleglength ///
        cervicaleheight trochanterionheight stature ///
        waistcircumference chestcircumference ///
        bicristalbreadth hipbreadth hipbreadthsitting
```

```
foreach v of local mmvars {
   replace `v' = `v'/10 if `v' < .
}

*******************************************************************************
// I. Create New Variables
// Continuous BMI

gen bmi = .
replace bmi = (weightkg) / ((stature/100)^2) ///
   if weightkg < . & stature < .

label var bmi "Body Mass Index (kg/m^2)"
format bmi %9.2f

// Categorical BMI
gen byte bmi_cat = .
replace bmi_cat = 1 if bmi < 18.5 & bmi < .
replace bmi_cat = 2 if bmi >= 18.5 & bmi < 25
replace bmi_cat = 3 if bmi >= 25 & bmi < 30
replace bmi_cat = 4 if bmi >= 30 & bmi < .

label define bmi_lbl ///
   1 "Underweight (<18.5)" ///
   2 "Normal weight (18.5-24.9)" ///
```

```
   3 "Overweight (25.0-29.9)" ///
   4 "Obese (>=30)", replace

label values bmi_cat bmi_lbl
label var bmi_cat "BMI category (standard clinical cutoffs)"

// Season Obs. Recorded

gen month_obs = month(date)

gen byte season = .
replace season = 1 if inlist(month_obs, 12,1,2)
replace season = 2 if inlist(month_obs, 3,4,5)
replace season = 3 if inlist(month_obs, 6,7,8)
replace season = 4 if inlist(month_obs, 9,10,11)

label define season_lbl ///
   1 "Winter" ///
   2 "Spring" ///
   3 "Summer" ///
   4 "Fall", replace

label values season season_lbl
label var season "Season measurement recorded"
```

```
// Body Types

/*******************************************************************

 Section 2e: Body type classification (sex-specific)

 - Inputs: stature (cm), bmi, waistcircumference (cm)

 - Derived: WHtR (waist-to-height ratio), sex-specific tertiles

 - Output: body_type (1-5) with value labels

*******************************************************************/


*-----------------------------------------------------------*

* 1) Waist-to-height ratio (central adiposity proxy)       *

*-----------------------------------------------------------*

gen whtr = .

replace whtr = waistcircumference / stature if waistcircumference < . & stature < .

label var whtr "Waist-to-height ratio (waist cm / height cm)"

format whtr %6.3f


* Common clinical threshold: WHtR >= 0.50

gen byte central_adip = .

replace central_adip = 1 if whtr >= 0.50 & whtr < .

replace central_adip = 0 if whtr < 0.50

label define central_lbl 0 "No central adiposity (WHtR<0.50)" 1 "Central adiposity (WHtR>=0.50)",
replace

label values central_adip central_lbl

label var central_adip "Central adiposity indicator (WHtR>=0.50)"


*-----------------------------------------------------------*
```

```
* 2) Sex-specific tertiles for stature and BMI            *
*-----------------------------------------------------------*
* 1=Low (short/lean), 2=Mid, 3=High (tall/heavy)
egen ht_tertile=xtile(stature), nq(3) by(gender)
egen bmi_tertile=xtile(bmi)if bmi < ., nq(3) by(gender)

label define tert_lbl 1 "Low" 2 "Mid" 3 "High", replace
label values ht_tertile tert_lbl
label values bmi_tertile tert_lbl

label var ht_tertile  "Stature tertile within gender"
label var bmi_tertile "BMI tertile within gender"

*-----------------------------------------------------------*
* 3) Map to 5 body types                                   *
*-----------------------------------------------------------*
gen byte body_type = .
label var body_type "Body type (gender-specific height/BMI + central adiposity)"

* Core corner types
replace body_type = 1 if ht_tertile==1 & bmi_tertile==1
replace body_type = 2 if ht_tertile==1 & (bmi_tertile==3 | central_adip==1)
replace body_type = 4 if ht_tertile==3 & bmi_tertile==1
replace body_type = 5 if ht_tertile==3 & (bmi_tertile==3 | central_adip==1)
```

```
* Everyone else defaults to Average

replace body_type = 3 if missing(body_type) & ht_tertile < . & bmi_tertile < .


label define bodytype_lbl ///

   1 "Small-Lean" ///

   2 "Small-Heavy" ///

   3 "Average" ///

   4 "Tall-Lean" ///

   5 "Tall-Heavy", replace


label values body_type bodytype_lbl


*******************************************************************************

// Section 2: Anthropometric Characteristics

*******************************************************************************


************************* Summary Table for Entire Sample ************************

// Summarize sample by body part

* Stats: n_obs, n_missing (%), n_suspicious (%), mean, min, q1, median, q3, max



// Define anthropometric variable groups

* Trunk circumference

local trunk hipbreadthsitting ///

                    waistcircumference chestcircumference ///
```

```
    bicristalbreadth hipbreadth



* Stature-related measures
local stature_grp span ///
                    stature kneeheightmidpatella ///
    cervicaleheight trochanterionheight ///
    waistheightomphalion functionalleglength ///
    footlength thumbtipreach



// Summary statistics table
dtable `trunk' `stature_grp', ///
  continuous(`trunk' `stature_grp', ///
  stat(n mean sd median min max)) ///
  title("Table 3.1-1. Anthropometric Measures (cm)") ///
        column(summary( N Mean (SD) Median Min Max))



// Output to word
collect style putdocx, layout(autofitcontents)
collect export extable2.docx, replace

************************ Hip-Height / Total Height Pct ************************
```

```
// Create hip height pct variable
gen pct_hip_height = .
replace pct_hip_height = (trochanterionheight / stature) * 100 ///
   if trochanterionheight < . & stature < .

label var pct_hip_height "% of total height attributable to hip height"
format pct_hip_height %6.2f

// Tabulate by sex
asdoc tabstat pct_hip_height, by(gender) stat(mean sd p25 p50 p75) save(pct_hip_height_tbl.doc)

// Scatter plot by sex
twoway (kdensity pct_hip_height if gender=="Female") ///
     (kdensity pct_hip_height if gender=="Male"), ///
     legend(label(1 "Female") label(2 "Male")) ///
     title("Fig 3.1 Distribution of Hip Height as % of Total Height") ///
     ytitle("Density") xtitle("Percent of Total Height")

*******************************************************************************
// 3. Data Quality Assessment
*******************************************************************************
*******************************************************************************
* 3A. Missingness table (numeric variables only; string missings excluded)
* Output columns (as strings for clean Word export):
*   Total Non-Missing (%), Total Missing (%), .a (%), .b (%), .c (%), .d (%), .e (%)
```

```
*******************************************************************************

* Identify numeric variables (these can contain .a–.e)
ds, has(type numeric)
local numvars `r(varlist)'

* Total observations in current analytic dataset
quietly count
local N = r(N)

* Build a small results dataset using postfile
tempfile miss_tbl
tempname posth

postfile `posth' ///
  str32 varname ///
  str20 nonmiss str20 miss ///
  str20 miss_a str20 miss_b str20 miss_c str20 miss_d str20 miss_e ///
  using `miss_tbl', replace

foreach v of varlist `numvars' {

  * Counts
  quietly count if !missing(`v')
  local n_non = r(N)
```

```
quietly count if missing(`v')

local n_mis = r(N)


quietly count if `v' == .a

local n_a = r(N)


quietly count if `v' == .b

local n_b = r(N)


quietly count if `v' == .c

local n_c = r(N)


quietly count if `v' == .d

local n_d = r(N)


quietly count if `v' == .e

local n_e = r(N)


* Percents of total N

local p_non = 100*`n_non'/`N'

local p_mis = 100*`n_mis'/`N'

local p_a   = 100*`n_a'/`N'

local p_b   = 100*`n_b'/`N'

local p_c   = 100*`n_c'/`N'
```

```
  local p_d   = 100*`n_d'/`N'
  local p_e   = 100*`n_e'/`N'

  * Post formatted cells: "count (pct%)"
  post `posth' ///
    ("`v'") ///
    (string(`n_non',"%9.0f")+" ("+string(`p_non',"%4.1f")+"%)") ///
    (string(`n_mis',"%9.0f")+" ("+string(`p_mis',"%4.1f")+"%)") ///
    (string(`n_a',"%9.0f")+" ("+string(`p_a',  "%4.1f")+"%)") ///
    (string(`n_b',"%9.0f")+" ("+string(`p_b',  "%4.1f")+"%)") ///
    (string(`n_c',"%9.0f")+" ("+string(`p_c',  "%4.1f")+"%)") ///
    (string(`n_d',"%9.0f")+" ("+string(`p_d',  "%4.1f")+"%)") ///
    (string(`n_e',"%9.0f")+" ("+string(`p_e',  "%4.1f")+"%)")
}

postclose `posth'

* Export to Word
preserve
use `miss_tbl', clear

putdocx begin
putdocx paragraph, halign(left)
putdocx text ("Table 3 Missingness by variable")
```

```
putdocx table t_miss = data("varname nonmiss miss miss_a miss_b miss_c miss_d miss_e"),
varnames

putdocx save "missingness_table.docx", replace

restore


*******************************************************************************

* 3B. Flag tallies table

* Output columns:

*   flag_name | sum(flag==1) (% of total N)

*******************************************************************************


* Total observations in current analytic dataset

quietly count

local N = r(N)


* Identify flag variables (edit patterns if needed)

ds flag_* dup_* keep_best flag_possible*, has(type numeric)

local flagvars `r(varlist)'


tempfile flag_tbl

tempname postf


postfile `postf' ///

   str32 flag_name ///

   str20 flagged ///

   using `flag_tbl', replace
```

```
foreach f of varlist `flagvars' {

   * Count flagged (assumes flags are 0/1; treats missing as not-flagged)
   quietly count if `f' == 1
   local n_flag = r(N)
   local p_flag = 100*`n_flag'/`N'

   post `postf' ///
      ("`f'") ///
      (string(`n_flag',"%9.0f")+" ("+string(`p_flag',"%4.1f")+"%)")
}

postclose `postf'

* Export to Word
preserve
use `flag_tbl', clear

putdocx begin
putdocx paragraph, halign(left)
putdocx text ("Table 3.Y. Data-quality flags (N = `N')")

putdocx table t_flags = data("flag_name flagged"), varnames
putdocx save "flag_tallies.docx", replace
```

```
restore
*******************************************************************************
// 4. Analysis
*******************************************************************************


// 4.1. correlation between  measures of stature


* Stature-related measures
local stature_grp span ///
                    stature kneeheightmidpatella ///
    cervicaleheight trochanterionheight ///
    waistheightomphalion functionalleglength ///
    footlength thumbtipreach
```